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Control of Continuum Robots for Medical Applications: State of the Art

M. T. Chikhaoui, J. Burgner-Kahrs
Laboratory for Continuum Robotics, Leibniz Universitaet Hannover, Hanover, Germany

Abstract:
Continuum robots are constructed from flexible materials, possess slender shapes, and can perform complex 3D motion. With their miniaturization capacity, high dexterity, and lack of rigid-links, their use in medical applications is increasingly considered. The specificity of their scale, their model differences, as well as the intended application requirements induced the developments of multiple control approaches. This paper provides an overview about control strategies of continuum robots and focuses particularly on those considered for medical applications. Furthermore, technical requirements such as sensor choices and actuation systems, as well as theoretical foundations including modeling approaches and autonomy levels are discussed in the scope of control of continuum robots. Finally, this paper elaborates on current challenges and open questions.

Keywords: Continuum Robots, Soft Material Robots, Control, Medical Applications, Miniaturized Robots.

Introduction

Continuum robots achieved a breakthrough in robotics over the last decades and opened the scope for novel approaches requiring navigation and compliance in constrained environments. Typically, in applications where standard serial or parallel robots with rigid links do not comply to geometric or interaction requirements, continuum robots represent a suitable alternative. Indeed, they can be highly miniaturized (down to 2 mm in diameter), describe complex shapes in 3D, and exhibit intrinsically high flexibility. As it can be seen from the examples in ***Fig. 1***, continuum robots do not include rigid links in their backbone, which describes *curves with continuous tangent vectors* [1]. These are defined in contrast to hyper-redundant robots where the number of joints/degrees of freedom (DoF) is large [2].

Due to their particular structure, research in continuum robots includes important challenges in design, fabrication, modeling, and sensing [3, 4, 5, 1, 6]. The choice of the materials, their compatibility with the environment, their nonlinear behavior, as well as their miniaturization raise ubiquitous research questions since the initial prototypes. While advancements have been made regarding design and modeling, a wide range of control approaches have been applied, yet without a particular convergence to a specific method. In conventional serial and parallel robotics, the scientific community found consensus regarding design (rigid links), actuation (defined joints as prismatic, revolute, cylindrical, etc.) and modeling (e.g. Denavit-Hartenberg parameters). This unification provided a matured and common starting point for the development of control strategies. Conversely, the non-uniformity of actuation methods, differences in kinematic structures, modeling approaches, and specificity of sensor integration in continuum robots prevented the development of universal control approaches thus far.

Fig. 1 *Examples of miniaturized continuum robots. a) Concentric tube continuum robot actuated by axial rotation and translation of pre-curved tubes (outer diameter 3 mm). b) Hybrid continuum robot with a combination of tendon-driven segments in a concentric assembly (outer diameter 8 mm). c) Tendon-driven continuum robot with extensible segments, using self-repelling magnetic spacer disks (outer diameter 8 mm).*

Paradoxically, while challenges regarding control of soft continuum robots were identified in the literature [7, 8, 9, 10], few research is clearly focused on control. Indeed, to the question “What aspects of soft robotics are you investigating (max 3)?” only 9% of the researchers answered “Control” [11].

Monitoring and influencing the behavior of continuum robots is addressed in the literature on a design optimization level (predefined) and/or motion planning level (precomputed), with the assumption that the physical system will not deviate from the

modelled/estimated behavior. However, intensive experimental assessments invalidated this assumption as accurate modeling is relative and hardly accounts for variable/dynamic environment. This paper discusses state of the art methods and technologies used for online control of continuum robots. We provide a comprehensive overview of control strategies used in this continuously growing field of research and guidance to the current challenges and future directions.

Preliminaries

In general, a robotic system responds to modifications in the actuators (input) by changes of its state (output). Control of a robot is concerned with finding the suitable actuation to reach a desired state, sought after to execute a given task. The state of a continuum robot can be described by (i) end-effector position/pose, (ii) robot's shape, (iii) isolated forces and stiffness, (iv) distributed forces and stiffness, and (v) kinematic performances. The current research in control is mainly focused on the end-effector level with position/pose control, force control – considered in previous surveys – or recently with the enhancement of the kinematic performances [12]. However, mastering a continuum robot's body is mainly addressed at the actuation and/or design optimization levels. Current on-line/real-time controllers of the full shape are rather sparse, as well as control of distributed forces a continuum robot can apply along its length. The same assessment is valid for stiffness/distributed force control (except for stiffness tuning using tendons and springs [13]) or for improvement of kinematic performances along the body of continuum robots with multiple segments. For a better access and monitoring of their state, it is worth mentioning that one of the key features of continuum robots is their kinematic redundancy. Hence, the remaining DoF to perform the desired task should be appropriately used to address control problems concerning the robot's full body.

The challenges regarding continuum robots' modeling include both forward and inverse kinematics. The high material non-linearity and the numerous physical phenomena involved during their motion (torsion, friction, shear, extension, etc.) limit the forward kinematics' accuracy and extend the computation time [14, 3, 15, 16, 17, 18, 19, 20]. With this limitation, a closed-form solution of the inverse kinematics is an open research question. The current approaches rely on look-up tables [21, 22, 23], numerical approximations [24], or differential inverse kinematics [19, 25].

For practical use in medical applications, continuum robots are generally teleoperated. This allows the surgeon to control the robot motion at any time. In other terms, a user actuates an input device and the motion is transferred to the robot's end-effector via suitable control algorithms. Through the user's visual perception, eventually aided by images (cameras in industrial applications, intraoperative images in medical scenarios), the state of tasks can be estimated and the input motion is corrected accordingly by the user. In continuum robotics, teleoperation is performed in medical applications [26, 27, 28, 1, 29], with additional virtual fixtures to increase feedback quality and interpretation [12]. These additions contribute to increasing of the levels of autonomy of robots in general [30]. Including autonomy to migrate from direct control (e.g. teleoperation) to shared/supervised autonomy will ultimately provide with better motion quality and repeatability of tasks, as well as reducing duration in order to permit the user to optimize time for making more complex decisions and interpretation of complex situations.

Current approaches on continuum robot control are divided into open-loop control, closed-loop control, or the combination of both. One of the initial researches focused on the comparison of the aforementioned control methods for a tendon-driven catheter. Open-loop control is shown to be highly dependent on the initial state and configuration of the continuum robot, resulting in low trajectory tracking accuracy. However, closed-loop control shows better performances in resolved motion rate [31]. This assessment is valid for other continuum robot architectures where model inaccuracy and omission of dynamics are prevalent. The following sections discuss examples of both controllers.

Open-Loop Control of Continuum Robots

Open-loop control essentially relies on model inversion in order to find suitable actuation values to reach the desired state. This approach is particularly useful if sensor integration is not available or desirable. Therefore, accurate modeling is one of the main challenges in continuum robotics. In order to achieve a satisfactory description of the shape, often complex models need to be derived to account for the numerous physical phenomena undergone by the robot. With the use of sophisticated non-linear models (i.e. accounting for non-constant curvatures), the challenge of inverse kinematics – required for open-loop control – rises.

In order to increase model accuracy and to include contact forces, researchers focus on finite element modeling. To this end, a model is validated on a

prototype of a tendon-driven soft continuum robot via visual feedback [32]. The inversion of the model is based on an optimization algorithm formulated as a quadratic programming problem with linear complementarity constraints (see [33] for further details). A different data-driven approach is conducted to solve the inverse statics of a comparable non-redundant continuum robot [34]. The accuracy and computation time of the proposed supervised learning (neural network) approach outperform a Jacobian approach in planar study cases. This is due to model inaccuracies resulting from additional friction, precision of fabrication, and actuation cable discrepancy.

The previous examples show that model inversion and accuracy represent the main challenges regarding open-loop control of continuum robots. Often, the complexity of finding closed-form solutions requires the introduction of simplifying assumptions such as non-redundancy, planar cases, or single segment continuum robots. Although data-driven methods exhibit good performances, the learning phase duration and the initialization process limit their application to general continuum robots with multiple DoF. While implementation of open-loop control methods is technologically compatible with medical applications (insofar the continuum robot is compatible), the exhibited performances in terms of accuracy, robustness, and computation rate need to be improved to increase the safety for clinical integration.

Closed-Loop Control of Continuum Robots

While open-loop control schemes highly rely on model inversion, closed-loop control includes feedback of robot's actuator and/or state. Both the measured error and the model contribute to reaching the desired state. Therefore, closed-loop control uses the reliability of the measured data to compensate for model inaccuracy during control. Methods of feedback control of continuum robots are discussed in the following.

Differential inverse kinematics are commonly utilized in closed-loop control of continuum robots and include the use of electromagnetic sensors for position feedback with estimated Jacobian [35] or analytically computed [36, 28, 37, 38], as well as the use of visual feedback, with an eye-to-hand configuration [39, 40] or eye-in-hand configuration [41], among other feedback techniques. This control approach is suitable for resolved rate motion, as the robot's Jacobian is locally defined in the given configuration using the forward kinematics.

The derivation of dynamics of continuum robots with multiple DoF is still an open research question. Ideally, the availability of such models opens the scope for applying well-established automatic control theories for serial [42, 43] and parallel robots [44]. Continuum robot dynamics in 3D space are derived using circular arc assumption for multiple segments [45, 46, 47], Hamilton's principle [48, 49], or including torsion, shear, and external forces to single-segment soft continuum robot [50]. However, their use in robust control is only introduced in few examples. These include a 9 DoF planar continuum robot controlled both in configuration and in task space [51]. Further, a 4 DoF underactuated tendon-driven soft robot, controlled using partial feedback linearization (to overcome nonlinear dynamics) and passivity-based controller (to overcome closed-loop control dynamics) [52], or via fractional order control (to avoid empirically tuned gains) [53]. Regarding the challenge of estimation of model uncertainties and input disturbances, a robust controller using interval arithmetic is implemented [54]. Therefore, a controller that accounts for user-defined tracking performances is made possible and represents an alternative to the previous model-based approach.

In summary, closed-loop control schemes represent a safer approach than their open-loop counter-parts regarding accuracy, stability, and robustness if the environmental changes occur at low frequencies. However, feedback control raises major technological challenges in sensor integration in highly miniaturized continuum robots in general; and more particularly in medical applications, concerning compatibility with the anatomy as well as with the preoperative imaging modalities and surgical instruments.

Advanced control

In order to overcome model inaccuracies, promising approaches based on closed-loop schemes focus on adaptive control. One of the first approaches in this area assumes the inaccuracy of concentric tube robot's kinematics in general to be bounded and known [55]. When the uncertainty is unpredictable, model parameter estimation becomes essential in adaptive control. Therefore, another approach focuses on model parameter estimation based on feedback while using the same controller of concentric tube robots [56]. Nevertheless, the latter method is efficient in a same workspace region with a predefined trajectory. A different inaccuracy issue is addressed for a soft continuum robot with uncertain length and base frame due to contact/locking in the environment [57]. The authors propose to reduce the model complexity by linearization for parameter

estimation and derive an adaptive control approach from state of the art [58].

Data-driven approaches:

With the high complexity of continuum robot kinematics and required computation time, recent research focuses on “model-less” approaches as an alternative. In fact, the following research includes a minimal use of the model (for initialization) or a complete model-free approach. Initialization with a constant curvature model is performed in a closed-loop control scheme [59]. A completely model-free approach is proposed by wobbling each actuator independently then normalizing the obtained Jacobian matrix, with extended data selection [60] or with Kalman filtering [61] to reject motion artifacts. Model-free approaches show faster convergence time and better performances in constrained environment. This is particularly interesting when external disturbances cannot be measured/estimated or taken into account by the model.

To some extent, the described approaches can be considered as data-driven although the model knowledge is included at different levels. Control schemes based on fast-learned models (~ two hours) represent an interesting alternative when deriving accurate kinematics that account for external disturbances is challenging. This approach is enhanced by a combination with continuous update of the robot’s model using end-effector feedback in order to adapt the motion in unstructured environments [62]. A comparable approach is introduced using a two-level inverse-kinematics solver. The first relies on gradient descent from task space to configuration space, and the second learns the mapping between configuration space and actuation space via neural network [63]. While data-driven methods show comparable results to model-based approaches, there is still no consensus regarding the learning approach. A comparison of three regression methods (neural networks) for inverse kinematics that outperform traditional inverse kinematics is conducted [64]. The compared regression methods are extreme learning machine, Gaussian mixture regression, and K-nearest neighbors’ regression. While the latter method shows better tracking performances, the robustness of the controller is not assessed. The proposed learned inverse kinematics could ultimately serve as a feedforward controller to predict feasible inputs and could be combined to a feedback loop to increase the system’s robustness.

One the one hand, data-driven approaches are beneficial in changing environments/external disturbances and variable behavior of continuum robots due to their flexibility. Furthermore, such methods can reduce the need for complex mathematical modeling approaches and can contribute to the popularity of continuum robots. On the other hand, high computation time, low compatible rate of change in the environment/disturbances, as well as complexity of learning approaches (due to non-linearity and redundancy of continuum robots) may prevent their use in medical scenarios. The adaptivity with different continuum robotic structure represents an additional challenges to learning methods, due to the specificity of each architecture and the non-uniformity of actuation and model descriptions.

Two-level controllers:

Further research on control focuses on pneumatically actuated soft continuum robots. Actuation uncertainties (pneumatic and mechanical), structure memory, friction, and hysteresis of the studied continuum robot prevent the uniqueness of the model solutions [65, 66]. Indeed, the soft continuum robot exhibits a non-stationary behavior (i. e. different static positions for the same actuation) due to pneumatic actuation. Therefore, its kinematics can neither be injective or surjective, which highly limits the applicability of standard control schemes. Then, a two-level control scheme is proposed based on neural network. The first level ensures convergence of the Cartesian errors using learned inverse kinematics without considering non-stationary behavior. The second level drastically reduces the amount of required learning data by adapting the initial model at every iteration through back-propagation of the measured Cartesian errors through another dedicated neural network.

For comparable reasons, a two-level controller of pneumatic-actuated soft robots is investigated. One example is based on a high-level control in terms of selection of the actuation lines and bending sensors, combined to a low-level control of every component soft module [67]. A similar approach is applied to select the suitable motion patterns (elongation, contraction, bending, torsion, etc.) at a higher level, while a low-level controller achieves the selected motion, using visual feedback [68]. While this approach can drastically reduce the complexity of a redundant/highly actuated system, its generalization to arbitrary motions is challenging as the motion decomposition and mapping to the suitable actuation strategy might be complex and provides multiple solutions.

In summary, two-level controllers are suitable to meet specificity of different actuation technologies or to overcome the complexity of a hybrid model-

based/data-driven approach. However, complementarity and compatibility of the decomposed tasks for two-level controllers is required and needs to be proven beforehand, therefore limiting the generalization of this approach. Thus, both specific medical/surgical gestures and compatible continuum robot architectures should be identified in the future to benefit from cascaded control schemes. Extensive efforts in design and motion analysis of the robot, as well as pre-computation and decomposition of the intended tasks are required for a potential implementation in medical applications.

Combination of feedforward and feedback control

In order to overcome model complexity and compatibility of continuum robot control in unstructured environments, an alternative approach to two-level controller is the combination of feedforward (based on model knowledge and robot behavior) and feedback (based on sensor feedback and external disturbances) control.

Combining model knowledge and robustness of closed-loop control is originally applied to control the OctArm manipulator's length [69]. To compensate for the unknown parameters of the dynamic model, a feedforward term based on neural network is added to a standard feedback controller. The benefits of such combination is further assessed in an original robotic structure combining flexible continuum elements and standard rigid-links [70]. This approach is effective in handling the different behavior of the rigid and flexible elements such as linearity, accuracy, velocity, and range of motion using a high/low-frequency partitioning.

The contributions of both components in the latter examples are equal in terms of priority (parallel control). A different, cascaded approach is further applied to joint-space control of a soft humanoid robot's member. With the help of a state estimator (joint angle estimation) and a dynamic model, a model predictive controller is developed to provide commands for pressure actuators. The latter are then controlled in a closed-loop scheme using the same feedback, while the former controller minimizes a cost (in this case, position error) across a discrete finite time horizon [71, 72]. A nested open-loop controller inside a differential inverse-kinematics based closed-loop control scheme is investigated [73, 74]. The open-loop controller is based on a quasi-static inaccurate finite element model and allows to compensate for image feature losses during visual servoing. To meet the specificity of the extensible pneumatically actuated continuum robot dynamics, a cascaded control scheme is introduced [75]. Using feedback linearization and a combination of feedforward control and linear PD-controllers, an actuator-space controller is developed to control actuator length in the outer loop and pressure in the inner loop. This combination (feedback and feedforward) is further investigated and the benefits of each approach are assessed [76]. The required feedback control for reducing the positioning error is in turn altering the designed stiffness of soft continuum robots. Therefore, it is shown that a feedforward term introduces an anticipatory behavior, which tends to conserve the robot's stiffness.

Combining feedforward and feedback control is a promising approach as it allows dealing with complex robot behavior (low and high dynamics) and preserving their advantages in terms of active compliance. Concurrently, further research needs to be provided to ensure taking advantage of both approaches and to prevent accumulating disadvantages of (i) lower accuracy and high computational cost of open-loop control and (ii) limited dynamics and required sensory system for closed-loop control. This is particularly relevant regarding constraints in medical applications in terms of high accuracy requirements (feedback control), and minimization of trauma (feedforward control). Additionally, such approaches require that compliance of continuum robots can be actively monitored and actuated, which imposes further challenges in sensing and state estimation.

Multi-Task Control of Robot's Shape

In order to access a higher level of controllability over the state of continuum robots and to benefit from their kinematic redundancy, recent research focused on achieving multiple tasks (simultaneous or sequential) with the same robot. In terms of positioning tasks, a parallel control architecture is applied to a highly articulated continuum robot with a 6-DoF movable base. A weighted controller of individual points along the robot's length is implemented. Using the high redundancy, the closed-loop controller performed positioning of the end-effector and avoidance of obstacles with particular segments of the robot. This control scheme might interrupt one of the assigned tasks in case of interference with the supplementary task requirements, and may lead to failure in achieving one or both tasks [77, 78]. To overcome this problem, prioritization of tasks is suggested for redundant continuum robots. This approach is essentially inspired by null-space motion from serial robotics theory (for a details, please refer to [79, 80, 81]).

Preliminary results are obtained to control a planar hyper-redundant robot to conform to constrained workspace by controlling successive backbone points from the end-effector to the base [82]. In continuum robotics, hierarchical tasks are applied as first attempt to solve the inverse-kinematics of concentric tube robots [83]. Further, tendon-driven continuum robots benefit from this approach in the plane for trajectory tracking with the end-effector as a primary task while avoiding moving obstacles with the robot's body [84], and for spatial multi-segment robots to resolve coupling effects among the segments [28]. This approach is further developed for spatial kinematically redundant continuum robots. While the primary task is essentially to control the end-effector position/pose, the secondary tasks mainly included constraints in configuration space (in constant curvature assumption), and particularly the curvature of individual segments. In a first example, decreasing the curvature to reduce tension on the actuation tendons is defined as a secondary task [85]. In a second example, increasing the segment curvatures is preferred in order to reduce the end-effector occlusion based on an endoscopic view of a camera mounted near the robot's base [86].

Overall, hierarchical task control enables to include shape constraints, and ultimately opens the scope to perform proper shape control. The latter shape control is performed for a soft planar fluidic elastomer robot. It combines closed-loop control in configuration space (curvature and length of every component segment) using visual feedback. This is possible using a constant curvature model, with initial steps of motion planning to map the desired path into arc parameters [87, 88]. However, the generalization of such methods to more sophisticated models considering robot shapes with non-constant (general) curvatures is still an open research questions. To some extent, constant curvature assumptions hold in free-space (with generally lower accuracy). However, in the intended medical applications, continuum robots are generally in contact with the anatomy or with the access tools (e.g. if integrated into endoscope's channels) and are required to perform tasks and apply forces on tissues. Therefore, further multi-task control methods should be investigated to compensate for contact forces along the backbone while preserving accuracy at the end-effector level.

Force/Stiffness control

The aforementioned controllers are mainly concerning one of the main characteristics of continuum robots: their complex shape. However, one of the most attracting feature of continuum robots is their flexibility. Intensive research focuses on the flexibility from a design aspect, and advanced results are obtained for their actuation principles and strategies. Nevertheless, further efforts are required to take advantage of the flexibility of continuum robots by control. This concerns controlling the force they are able to provide and the stiffness undergone by the robot's backbone. Reaching remote zones and navigating in confined spaces is one of the main advantages of continuum robots in contrast to rigid-link robots (serial or parallel), however, application of specific tasks requires designing suitable force/stiffness controllers.

A hybrid force/position controller is investigated using visual feedback by approximating the continuum robot to a linear spring, and thus mapping the position to the generated force. The particularity here remains in the model-less approach, where the robot's Jacobian is estimated/learned via visual feedback. Contact estimation can be obtained by the use of force sensors and motion restriction in automatic control schemes [89] or based on end-effector tracking and comparison with model output in a teleoperation scheme [90].

Model-based approaches are also investigated to estimate and regulate the contact forces for continuum robots [91, 92]. Therefore, hybrid motion/force control is possible, and most of the approaches rely on prior work on task frame formalism [93, 94]. This approach is further investigated for continuum robots to achieve interaction control. The goal is to compensate for external disturbances at the end-effector level using both Jacobian and compliance matrices [95, 96]. The use of Jacobian and compliance matrices is further integrated to a multi-segment continuum robot with wrench estimation [97, 98]. A different approach based on deflection model is introduced to control the stiffness along the axis of the robot by transforming the stiffness constraints into task-space constraints [99].

In order to be able to control continuum robots for complex task-space motion and force application along the backbone, the current approaches can benefit from recent results about combination of feedback and feedforward control [76]. Multi-task control should also allow for synergy between the applied tasks and for avoiding task conflicts. With a unified modeling approach accounting for external forces, such approach have the potential to be generalized in continuum robotics. Nevertheless, with the technological challenge of sensor integration – especially in medical applications –, state

estimation (including distributed forces) should be sought.

Discussion

This review of the recent advances in continuum robot control permitted to identify major challenges inherently related to their particular morphology. These challenges concern the control of the state, with the shape in the first level. Indeed, multi-task control needs to ultimately include models that are more sophisticated to increase the accuracy. Moreover, this type of controller is required to integrate follow-the-leader constraints, as this is not generally achievable by any continuum robotic structure and necessitates intensive offline model-based motion planning [100, 101, 102, 103, 104, 105]. Such motion is desirable in order to allow for a minimal body coverage during navigation to conform to anatomical path, or if certain parts of the robot's body should be constrained to anatomical constraints while performing a task with the end-effector. These aspects require a suitable mapping of such additional tasks with respect to actuator space (i.e. in null-space motion). This is currently limited by the non-linearity of the models and the prevailing coupling between actuators and segments. Furthermore, sensing technology limitations prevent a full shape control from being effective, as a real-time shape sensing/estimation system is required. In fact, the availability of sensors for determination of the current state is an identified open research challenge, especially when the state includes more components than end-effector position/pose, such as distributed forces [106].

Furthermore, continuum robot's body control should also integrate distributed forces/stiffness. To this end, suitable descriptors of the robot's state and their mapping into actuator space are required. Nevertheless, this mapping should not alter the position/pose/shape control. An appropriate scheme consists in the balance and synergy between feedforward control, to keep the inherent flexibility of continuum robots, and feedback control, to provide more stability and accuracy in case of unstructured environment and unknown constraints. To overcome modeling inaccuracy, data-driven solution are proven to be efficient for continuum robot control. However, few examples show online application of model-less methods, and a general learning/estimation framework needs to be defined in the absence of a common modeling approach.

Such solutions are suitable for quasi-static motion. However, to benefit from the wide knowledge of current automatic control theory, dynamic modeling of continuum robots is required. This would open the scope for more advanced impendence controllers [107]. The latter controller will permit able to take full advantage of the inherent flexibility of continuum robots, in order to enable active stiffness, highly beneficial for the current applications where humans are required to interact with the robot. Whether it concerns industrial applications where the robot operates near humans or medical applications where the robot accesses the body and navigates in the anatomy, controlling distributed forces and stiffness along the backbone will be highly profitable.

Finally, (soft) continuum robots exhibit intrinsically non-linear behavior, as well as uncertainties due to non-negligible deviation from their mathematical models. Further optimal controllers may achieve a suitable stabilization of such systems, as non-linear H-infinity (H_∞) controllers, which allows setting pre-defined closed-loop performances, or combined with passivity-based approaches to overcome closed-loop dynamics [43]. Additionally, recent investigations on fuzzy logic control of continuum robots showed promising results in simulation to overcome model complexity of a single segment tendon-driven robot [108, 109]. However, added to experimental validation, such controllers require high computational efforts when deriving the appropriate gains and their generalization to other continuum robot architecture is yet to be proven.

In summary, control of continuum robots is a growing research area and requires joint efforts in both understanding/describing (mathematical modeling), monitoring (state estimation), and mastering (control theory) complex robot behavior. This observation was recently formalized in a global framework of morphological computation and control [110, 111]. First promising results concern morphological-computation based control of pneumatically actuated soft continuum robot [112]. Indeed, it is shown that traditional control, a morphological computation based approach, and active compliance can be combined to gain leverage in the control of the intrinsic features of continuum robots, and would ultimately lead to a safe and precise behavior.

For medical applications, whether it concerns mathematical models or learned counter-parts, description approaches should be made more accurate without increasing computation time to allow real-time applications. State estimation is closely related to modeling challenges, with additional restrictions when it comes to sensor integration due to miniaturization of medical continuum robots, as well as biocompatibility and accordance with medical images and tools.

Furthermore, in terms of control, multiple tasks and requirements need to be met, including task-space requirements such as accuracy and robustness to disturbances, compliance requirements such as passive and active compliance and force applications, as well as real-time requirements to update the suitable commands to fulfill the given tasks.

Conclusion

While design and modeling of continuum robots has matured over the last decades, no consensus has been made regarding a suitable control architecture heretofore. As reviewed throughout in this paper, many open research questions remain unsolved regarding the control of the state of continuum robots. While conventional control methods achieved satisfactory results in terms of end-effector control (pose, force), these approaches disregard the compliant nature of such robots and circumvent their advantages by assuming their behavior to be rigid.

Ultimately, an optimal control approach should take advantage of the flexibility of continuum robots to push forward the current applications and to include online navigation in cluttered, dynamic, and/or unstructured environments. Achieving such a high degree of controllability of continuum robots in terms of mastering their full state will open avenues of applications where the human is present. Examples include general human-robot interaction in industrial applications, or surgical operations and robot assistance as rehabilitation robotics. In this particular field, developing suitable control strategies will ideally increase the level of autonomy of medical continuum robots and improve the practitioner's experience during diagnosis and therapy. Recent results on rigid-link surgical robots show supervised autonomous soft tissue surgery, promising in enhanced efficacy, safety, and optimality of surgical techniques [113]. With the inherent safety and miniaturization of continuum robots, more reliable and less invasive autonomous medical tasks should be made possible. This approach can be further improved by augmenting the observation quality using visual virtual fixtures and haptic feedback [114, 115, 116].

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